PALM Seminar
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Information for participants

15-19 September 2025, Hannover, Germany

Dear PALM seminar participant...

thank you for joining the PALM seminar from September 15th to 19th, 2025. In the following, you will get all the important information you need for participating.

General remarks

The seminar is a **full-time seminar** with a workload of 8-9 hours a day. It consists of non-interactive lectures, exercises in the morning and hands-on sessions in the afternoon

To avoid false expectations, we remind you that the seminar purpose is a general introduction to the PALM model system for beginners, which also includes theoretical aspects of the Large-Eddy Simulation technique and the numerical methods that are used in PALM. In addition, we cannot address all the technical features and possible applications of PALM in just one week. The seminar provides the basic working knowledge to use PALM and develop own set-ups.

Venue

The seminar will take place in Hannover, which is the largest and capital city of Lower Saxony. For potential touristic activities visit https://www.visit-hannover.com/en

The seminar will be given in different rooms at or close to the Institute of Meteorology and Climatology (IMUK, https://www.meteo.uni-hannover.de/en/), which is a department of the Leibniz University Hannover (LUH). You can find the room numbers on the attached schedule. Lectures will be given at

Building 4107, Room 009 ("Kirchenkanzlei")

https://info.cafm.uni-hannover.de/en/building/4107,
https://info.cafm.uni-hannover.de/en/room/4107.000.009

and Exercises at

Building 4105, rooms F118 ("Seminarraum") and F139 ("Wetterraum")
https://info.cafm.uni-hannover.de/de/building/4105
https://info.cafm.uni-hannover.de/de/room/4105.001.F118
https://info.cafm.uni-hannover.de/de/room/4105.001.F139

Coffee breaks will provided in room F128 on the same floor.

Accommodation

We recommend to choose an accommodation close to the venue to limit the time you need for the daily arrival and departure as much as possible. In the following, some suggestions are given that are in walking distance to our institute or near the subway lines 4 and 5 which take you to the IMUK in less than 10 minutes:
https://www.hotel-in-herrenhausen.de/en/
https://www.hotel-am-herrenhaeuser-garten.de
https://werkhof-hannover.de/hotel
https://www.hannover-city-panorama.de/
https://hrewards.com/en/intercityhotel-hannover

Seminar schedule

Lectures | Room 009 | Hands-on | Room F118 & F139 | Coffee breaks & Ice Breaker | Room F128

	Monday	Tuesday	Wednesday	Thursday	Friday
8^{30}-10^{00} coffee break	Seminar Introduction & Fundamentals of LES Raasch	Numerics & Boundary Conditions Raasch Code Parallelization Raasch	Restart Runs, Non-Cyclic Boundary Conditions, User-defined code Gryschka	Land Surface Model (LSM) & Urban Surface Model (USM) Maronga	Bulk Cloud Model (BCM) Gryschka
10^{30}-11^{45} short break	LES for Beginners & PALM Overview Raasch	Code Structure & Features Raasch Topography Raasch	Grid Nesting Raasch	Radiation Models Schubert Dynamic Drivers Anders	Wind Turbine Model (WTM) Maronga
12^{00}-13^{00}	PALM Steering Raasch Data Visualization Raasch?	Debugging Raasch	Graphical User Interface: palmrungui Maronga	Static Drivers Schubert	E5: Wind Turbine Model Maronga
13^{00} -14^{30}	Lunch Break	Group Photo & Lunch Break	Lunch Break	Lunch Break	Lunch Break
14^{30}-15^{00}	E1: Convective Boundary Layer Maronga	E2: Neutral Boundary Layer Gryschka	E3: Flow Around Buildings Maronga	E4: Land Surface Model Maronga	E5
15^{00}-16^{00} coffee break	E1	E2	E3	E4	Discussion of Results/Problems Maronga
16^{30}-17^{00}	E1	E2	E3	E4	Final remarks / Closing Raasch/Maronga
17^{00}-18^{00}	Discussion of Results/Problems Maronga	Discussion of Results/Problems Gryschka	Discussion of Results/Problems Maronga	Discussion of Results/Problems Maronga	
18^{00}-21^{00}	Ice Breaker		Seminar Dinner		Date: 7/26/25

The lecture material is provided via the Leibniz University Hannover Cloud System called Seafile

https://seafile.cloud.uni-hannover.de/d/ed4a9c691a874d43966f/

It will be possible to download the provided PDF documents and PALM steering files **at latest one day before** the respective lecture or exercise is scheduled.

PALM data visualization

During the seminar, the script palmplot (after the download of palm located at packages/visualization/palmplot) will be used to plot the output data (netCDF) of the exercises. The script is based on the NCAR Command Language (NCL, https://www.ncl.ucar.edu/). Therefore, NCL has to be installed on your notebook/PC. On Linux systems you can install NCL via the software/package manager, e.g., for Ubuntu via

```
sudo apt install ncl-ncarg
```

If you have installed NCL correctly, the command "ncl" should be available in your terminal. In the recent past, we encountered several issues using NCL and the software is no longer maintained. Therefore, if you are not able to run palmplot properly or you prefer an alternative, we suggest to use **ncview** (https://cirrus.ucsd.edu/ncview/, also available via the package manager) or panoply (https://www.giss.nasa.gov/tools/panoply/). Also, please feel free to use any other alternative viewing/plotting software/library to visualize your PALM output (e.g., psy-view, matplotlib).

Social events

Each day, two **coffee breaks** are offered from 10:00 -10:30 CET and 16:00-16:30 CET. There, you can meet and exchange with other participants and have interesting discussions in-depth on or apart from the seminar topics. Snacks, cool and hot drinks will be provided.

On Monday, there is an **ice breaker** to get to know each other better and to establish some new contacts. Food and drinks are included. Everyone is welcome to participate!

Finally, an optional **seminar dinner** is planned for Wednesday evening. The costs are not included in the seminar fee. Moreover, you need to register to for the dinner until September 3 (mail to stenzel@meteo.uni-hannover.de). In case you have a special diet (vegan, vegetarian, etc.), please let us know.

PALM installation

To ensure a smooth start of the seminar, please install PALM (release 25.04) well in advance of the seminar. The lecture on PALM installation is available on the Seafile system via

https://seafile.cloud.uni-hannover.de/d/df858faf73c5432ead03/

The release 25.04 is available via
https://gitlab.palm-model.org/releases/palm_model_system/-/releases

Please note that there will be no time during the seminar to install PALM! Please ensure that PALM is installed and that the internal tests (see README.md in the repository after installation) is successful. If you encounter problems that cannot be solved (e.g. by your local system administrator or via the PALM installation document linked above), please contact us as soon as possible.

WiFi access during the seminar

Free WiFi in and around the venue is possible via eduroam (https://eduroam.org/). If you can not use this service, we will provide you temporary guest account for the whole 5 days of the seminar. Please ask for such an access during the seminar week.

Participation certificates

Every participant will receive a **PALM Training Certificate (Level 1+2)** after completion of the seminar. The certificates will be handed out on Friday afternoon. In case you need to leave early, please let us know in advance.

Contact (organizational support)

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Contact (technical support)

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